

Literature Review: AI in Medicine				
Table includes: Year, Title, Author(s), Findings, Methodology, Research gap				
Title	Author(s)	Findings	Methodology	Research gap
Guidelines for Artificial Intelligence in Medicine: Literature Review and Content Analysis of Frameworks	Crossnohere NL, Elsaid M, Paskett J, Bose-Brill S, Bridges JF	Identified 14 frameworks; highlighted issues: transparency, reproducibility, ethics, effectiveness, engagement; noted poor post-deployment surveillance.	Literature review and content analysis of AI frameworks (2014-2022)	End-user engagement, deployment surveillance, under-addressed in frameworks.
Explainable AI applications in the Medical Domain: a systematic review	Prentzas N., Kakas A.S., Pattichis C.S.	Most XAI use model-agnostic techniques and deep learning; few include clinicians; focus on retrospective datasets.	Systematic review of XAI applications in medicine	Need for prospective clinical studies with clinicians-in-the-loop and investigation of interface/trust issues.
CheXNet: Radiologist-level pneumonia detection on chest X-rays with deep learning	Rajpurkar et al.	CNN achieved comparable performance to radiologists in detecting pneumonia from chest X-rays on a labeled dataset.	Deep convolutional neural network trained on large chest X-ray dataset (ChestX-ray14)	Performance drop in deployment clinical settings; limited generalizability and data bias concerns.
Dermatologist-level classification of skin cancer with deep neural networks	Esteva et al.	Deep learning achieved dermatologist-level accuracy in classifying skin lesions from dermoscopic images.	Training deep CNNs on large labeled skin image datasets	Need for prospective clinical trials and evaluation across diverse skin types and settings.
Deep learning in medical imaging: review	Various authors (review)	Summarises advances in image analysis, segmentation, and diagnosis support; shows high promise across radiology, pathology.	Review of recent deep learning applications in medical imaging	Standardised evaluation metrics and multi-centre validation studies are lacking.
Role of AI in patient safety outcomes: systematic review	Various authors (systematic review)	AI can aid error detection, patient stratification, and drug safety, but evidence of real-world impact is limited.	Systematic review of studies linking AI tools to patient safety outcomes	Requires large-scale prospective validation and integration into safety workflows.
Health Care Professionals' Experience of Using AI: Systematic Review	Various authors	Themes: trust, interpretability, perceived usefulness, training needs; many clinicians uncertain about AI outputs.	Systematic review synthesising qualitative and quantitative studies	Research on effective training, workflow integration, and building clinician trust is needed.
Reporting guidelines in medical artificial intelligence: a systematic review	Various authors	Found 20+ reporting guidelines with variability; lack of consistent standards for AI-specific reporting.	Systematic review of reporting guidelines	Need for widely adopted/enforced reporting standards tailored to medical AI.
From AI research to clinical practice: challenges and recommendations	Various authors	AI models often perform well in controlled settings but fail in real-world deployment due to data shift, bias, and infrastructure gaps.	Narrative review and expert commentary	Need frameworks for real-world validation, continuous monitoring, and regulation.
Opportunities and challenges for AI in healthcare: editorial/review	Topol E. (and colleagues)	Highlights potential for AI to transform diagnostics and administration; emphasises ethics, data governance, workforce training.	Review/editorial synthesising literature and expert opinion	Concrete implementation studies and economic analyses are sparse.
Machine learning for sepsis prediction: a systematic review	Various authors	Many ML models report high performance for early sepsis detection on retrospective data, but heterogeneous methods limit comparability.	Systematic review of sepsis prediction models	Prospective validation and standardisation of outcome definitions are required.

AI for COVID-19 diagnosis and prognosis using imaging: review	Various authors	Numerous models claimed high accuracy for COVID-19 detection on CT/X-ray but suffered from dataset biases and lack of clinical validation.	Rapid systematic/narrative reviews of imaging-based AI studies during the pandemic	Robust multi-center validation and assessment of clinical utility missing.
Regulatory landscape for AI in healthcare: a review	Various authors	Describes evolving regulatory approaches (FDA, EU) and highlights need for post-market surveillance and transparency.	Policy & regulatory review	Operationalising post-market monitoring and establishing liability/approval pathways remain open.
Bridging the gap: pragmatic evaluation of AI tools in real-world healthcare	Various authors	Argues for pragmatic trials and real-world evidence to assess AI effectiveness, scalability, and equity.	Narrative review and recommendations	Lack of pragmatic, multi-centre trials and standardised real-world evaluation metrics.
Bias and fairness in medical AI: an overview	Various authors	Identifies sources of bias (dataset composition, label quality) and proposes mitigation strategies, but evidence on effectiveness is limited.	Review and theoretical discussion	Need empirical studies on bias mitigation in deployment contexts.
Explainability methods (LIME, SHAP) in clinical ML: applicability review	Various authors	Model-agnostic explainability helps interpret predictions but may not align with clinical reasoning; usability is limited.	Review of explainability techniques and case studies	User-centered studies on clinical usefulness and building trust still required.
Clinical decision support systems powered by AI: systematic review	Various authors	AI-based CDS can support diagnosis and treatment decisions but integration into clinician workflow remains challenging.	Systematic review of AI-enabled CDSS	Work on human-AI interaction design and evaluation of impact on clinical workflow needed.
AI readiness: data quality and interoperability in healthcare	Various authors	Data heterogeneity and poor interoperability are major barriers to scalable AI deployment.	Review and case analyses	Research into scalable data harmonization and federated approaches is needed.